

EXP 12: NB-IoT vs LTE-M Throughput Simulation Date

Objective 1 Matlab Code and Output

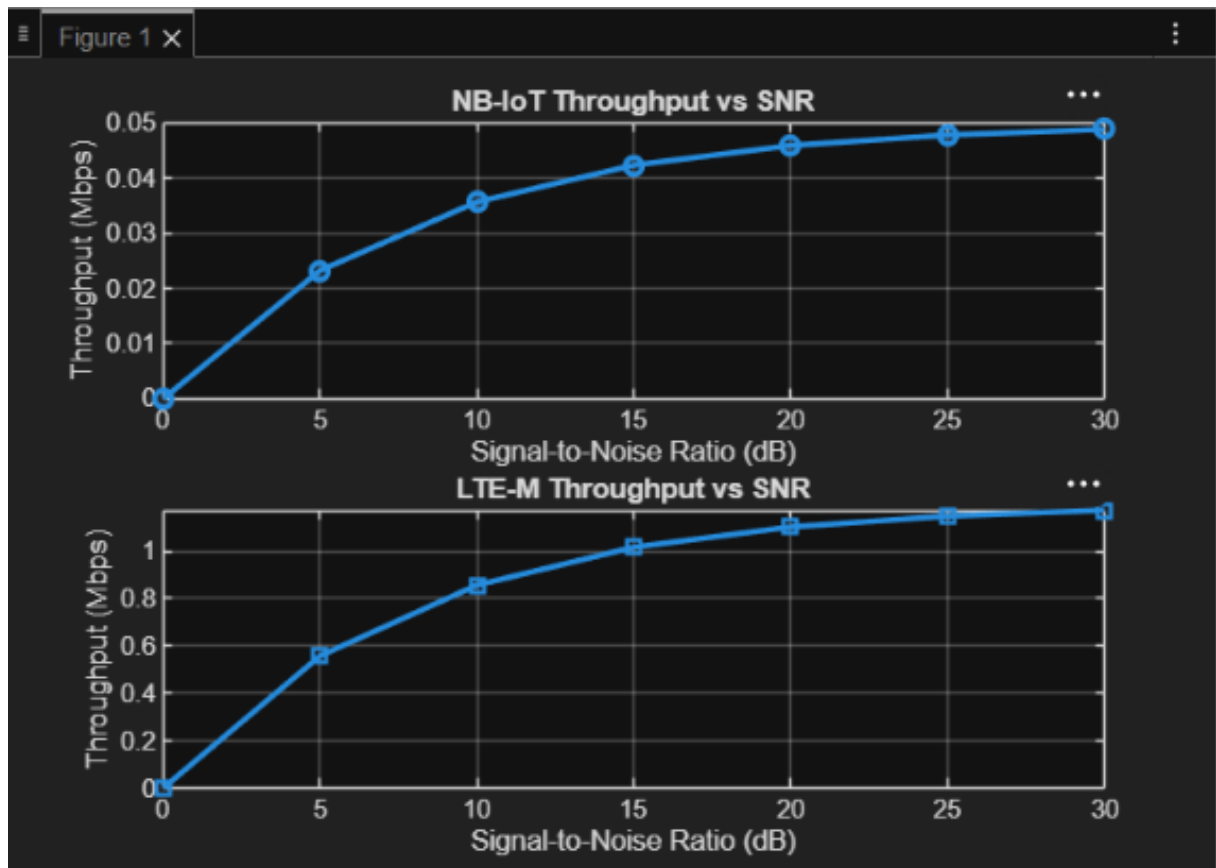
Program code:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M = 1.2*(1-exp(-SNR/8));
disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
disp([SNR' NB_IoT' LTE_M'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
```

Output:

Command Window		
SNR (dB)	NB-IoT(Mbps)	LTE-M(Mbps)
0	0	0
5.0000	0.0232	0.5577
10.0000	0.0357	0.8562
15.0000	0.0423	1.0160
20.0000	0.0459	1.1015
25.0000	0.0478	1.1473
30.0000	0.0488	1.1718
>>		



Objective 2 Matlab Code and Output

Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Latency (ms)
NB_IoT = [180 160 145 130 120 110 100];
LTE_M = [90 80 70 60 50 45 40];
disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')
disp([SNR' NB_IoT' LTE_M'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Latency vs SNR')

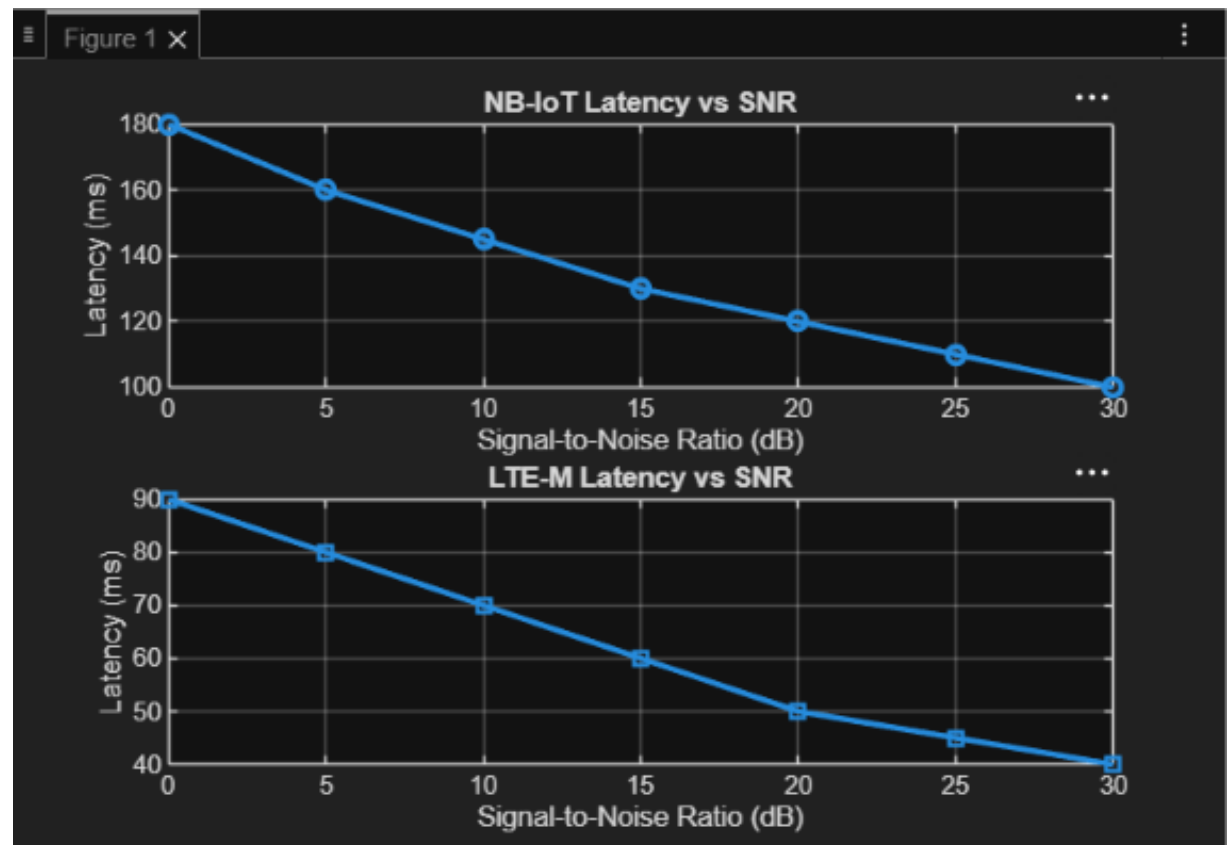
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Latency vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
```

Output:

Command Window		
SNR(dB)	NB-IoT Latency(ms)	LTE-M Latency(ms)
0	180	90
5	160	80
10	145	70
15	130	60
20	120	50
25	110	45
30	100	40

>>



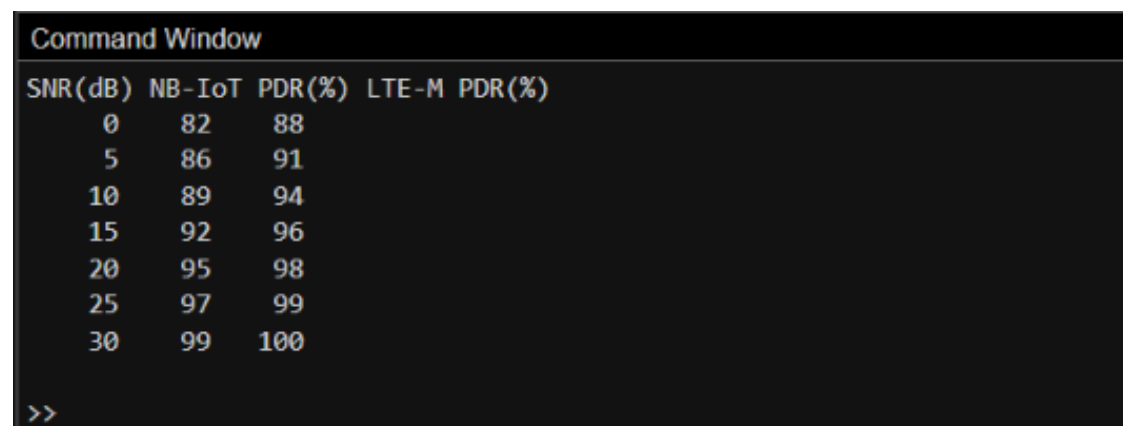
Objective 3 Matlab Code and Output:

Program code:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Packet Delivery Ratio (%)
NB_IoT = [82 86 89 92 95 97 99];
LTE_M = [88 91 94 96 98 99 100];
disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%)')
disp([SNR' NB_IoT' LTE_M'])

figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
```

Output:



SNR(dB)	NB-IoT PDR(%)	LTE-M PDR(%)
0	82	88
5	86	91
10	89	94
15	92	96
20	95	98
25	97	99
30	99	100

